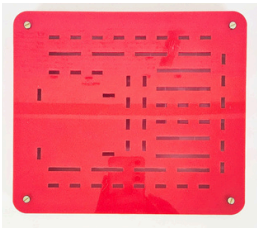
SCIENCEUTSAV®
Science is awesome

I'm a ROBO SCIENTIST

Touch-Based Smart Control Systems

Session 7

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


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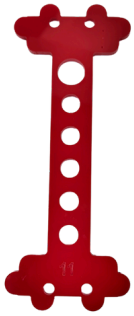


1

Spacers

2

L Clamp

3

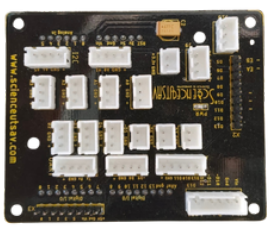
Spacer Screws

4

Nuts

5

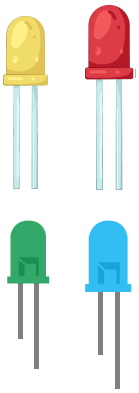
Screw

6

Arduino With Shield

7

Arduino cable

ELECTRONIC COMPONENTS

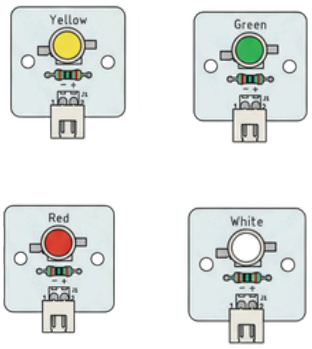


DIP LED (Dual In-line Package LED)

A DIP LED is a traditional through-hole light-emitting diode designed with two leads for easy mounting on printed circuit boards. It provides reliable and efficient illumination for indicators, displays, decorative lighting, and electronic projects, and is available in various colors, sizes, and brightness levels.

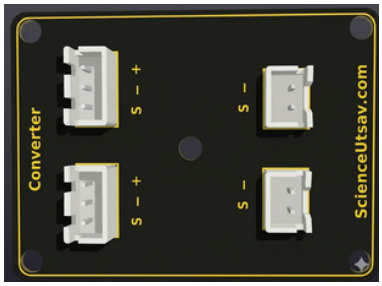
Touch Sensor

A touch sensor is an electronic input device that detects touch or proximity from a finger without requiring mechanical pressure. It provides a simple and reliable way to control electronic circuits, making it ideal for switches, automation systems, consumer electronics, and interactive projects.



1 Watt LED PCB

1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.



Converter

A converter is an electronic module that changes electrical power from one voltage level, current type, or signal format to another. It enables compatibility between different devices and power sources, making it widely used in power supplies, battery-operated systems, industrial equipment, and electronic projects.

String LED

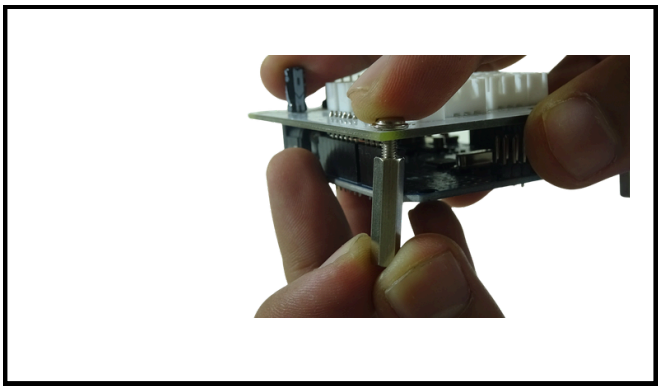
A String LED is a series of multiple LEDs connected along a flexible wire or cable to provide decorative and ambient lighting. It offers energy-efficient illumination and is widely used for home décor, festive decorations, events, signage, and creative lighting projects.

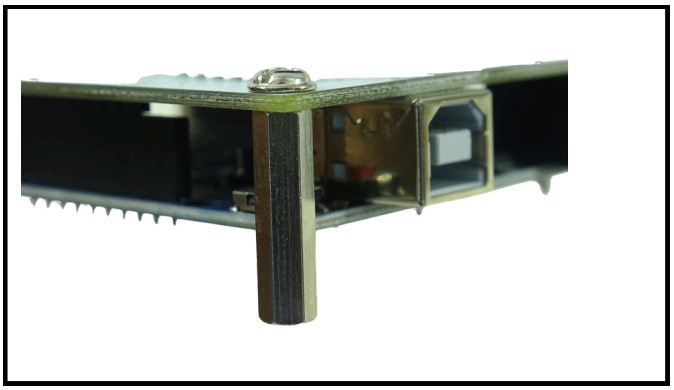
2 PIN RMC

Used to connect the 1-watt LEDs and the buzzer.

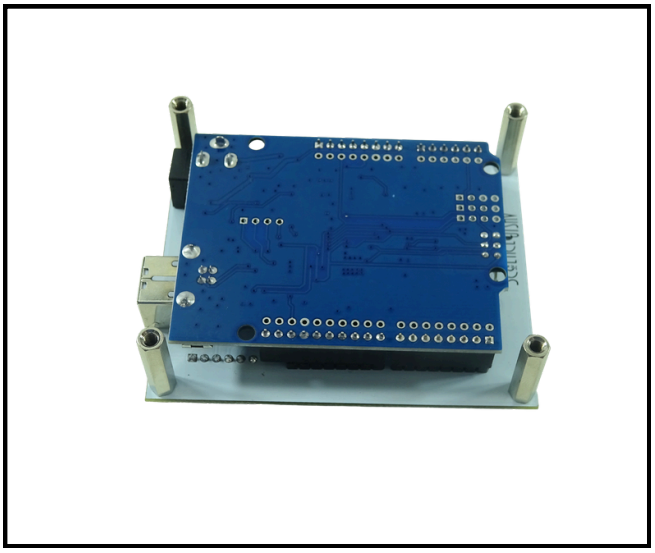
Procedure :

Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw

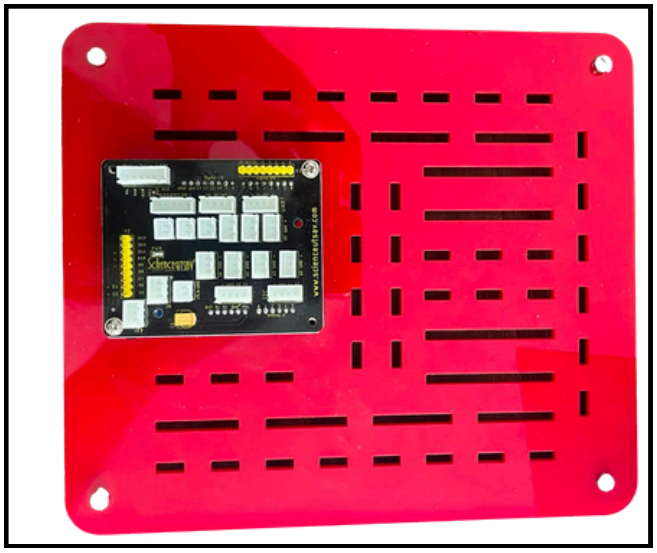




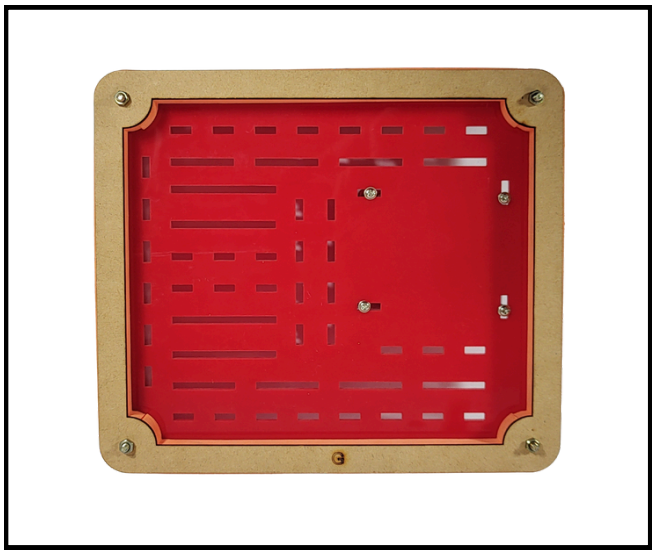
Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



Step 3 : Place the Arduino Shield to the base board as shown below

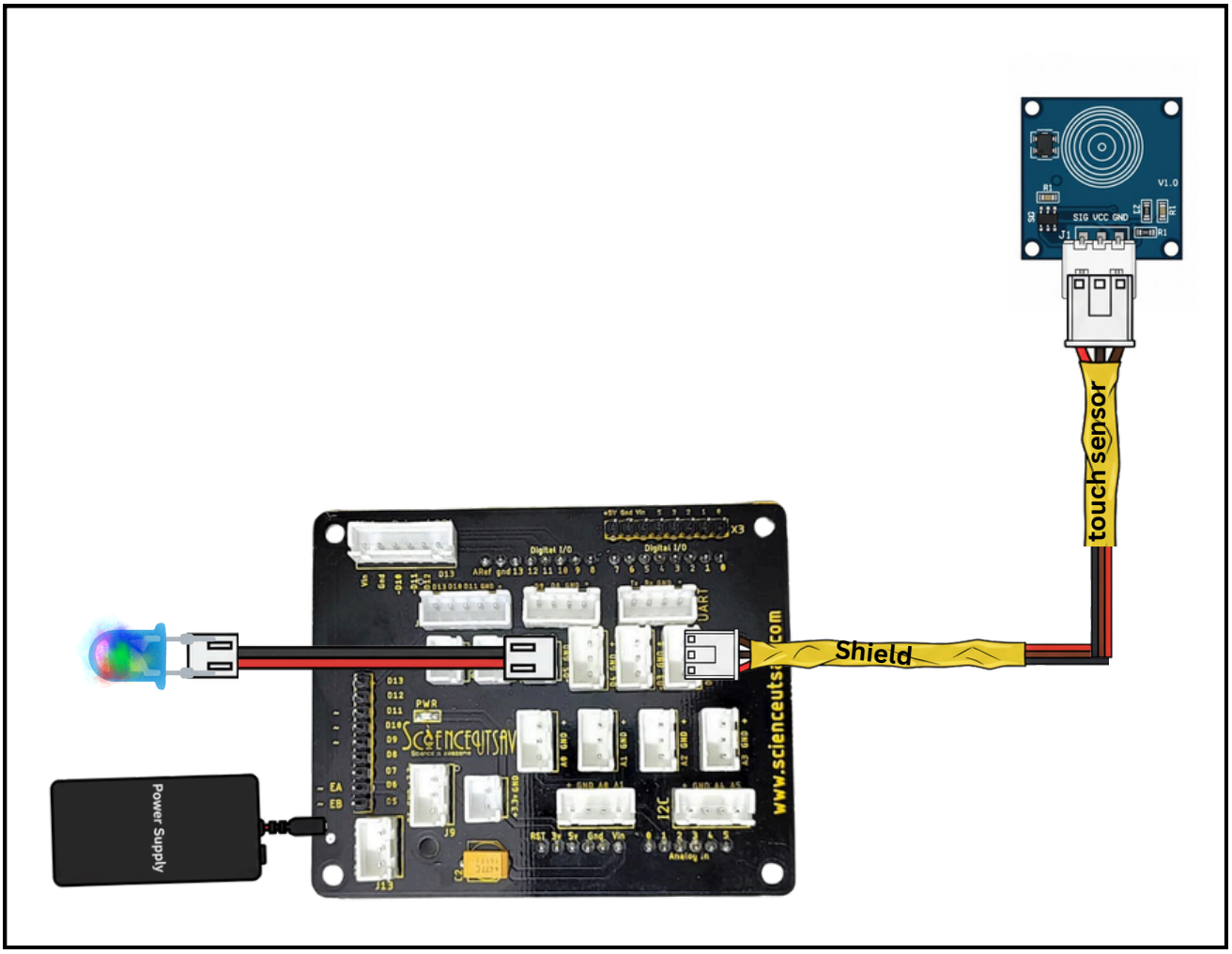


Step 4 : Fix it with the spacer screw from the bottom as shown below



BEGINNER PROJECT

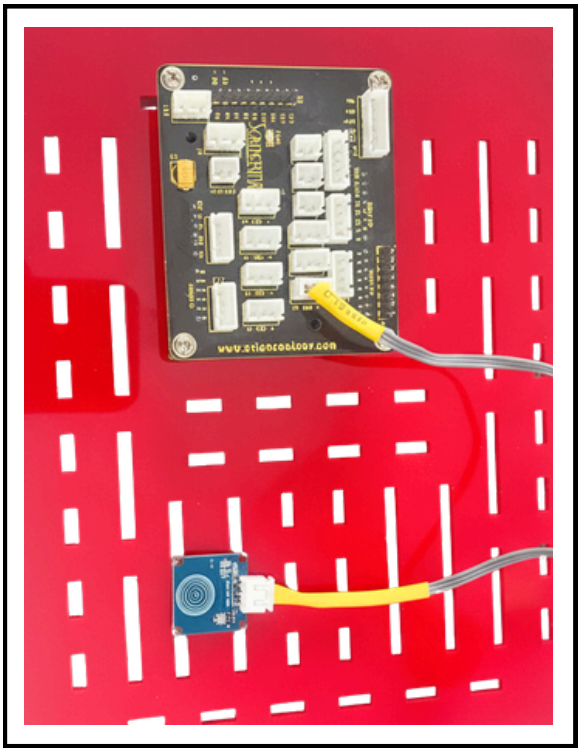
The circuit diagram of the project is given below



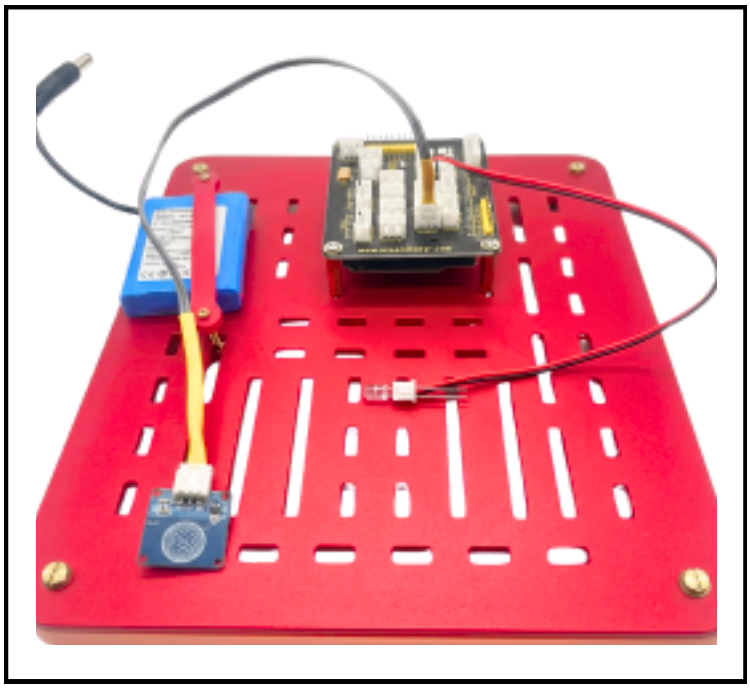
CONNECTIONS:

1. TOUCH SENSOR - D2
2. MULTICOLOUR LED - D5

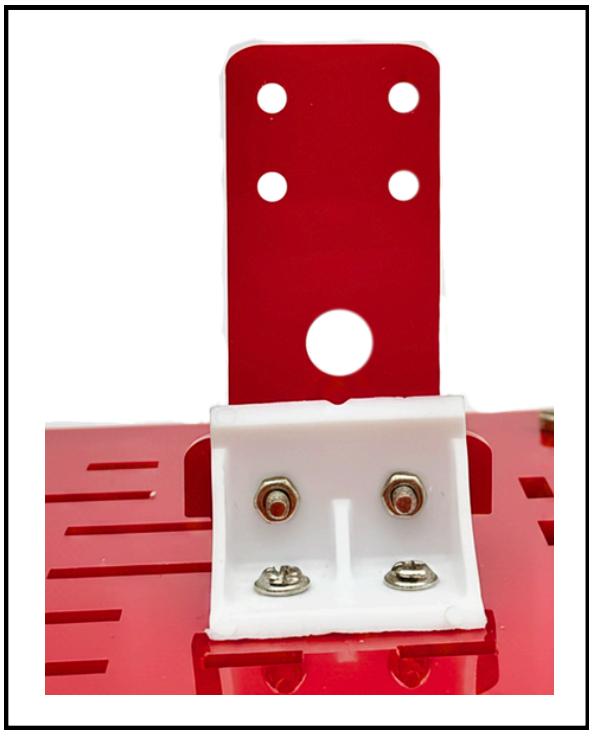
Step 1- Attach the touch sensor to the aurdino as shown



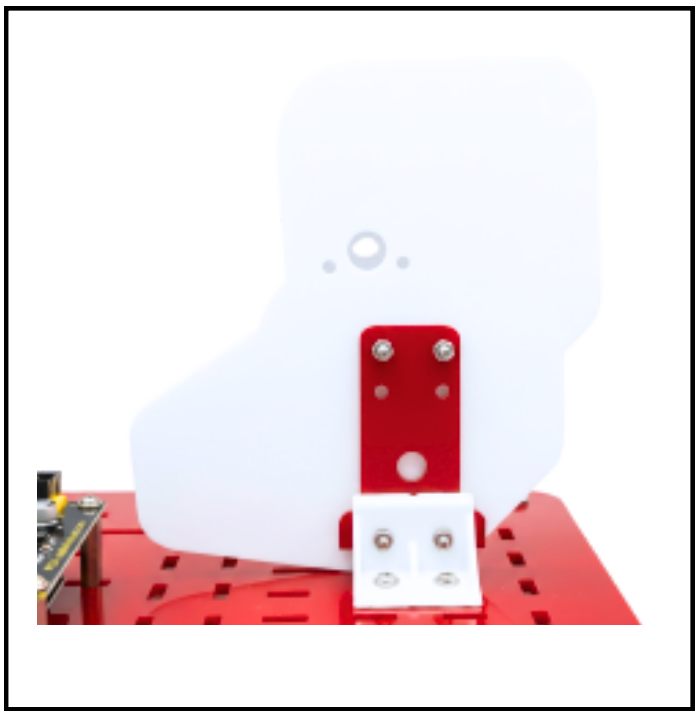
Step 2- Connect the dip LED to the aurdino



Step 3- Attach the stand as shown with the help of L-clamp

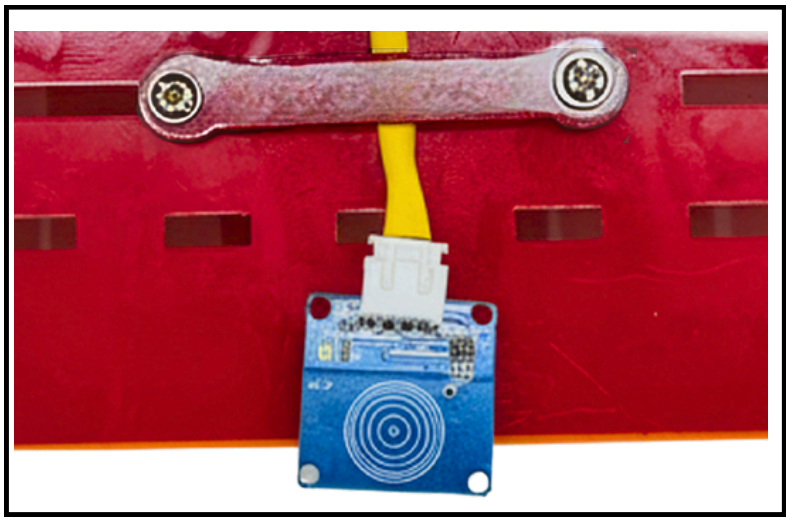


Step 4- Attach the template to the stand

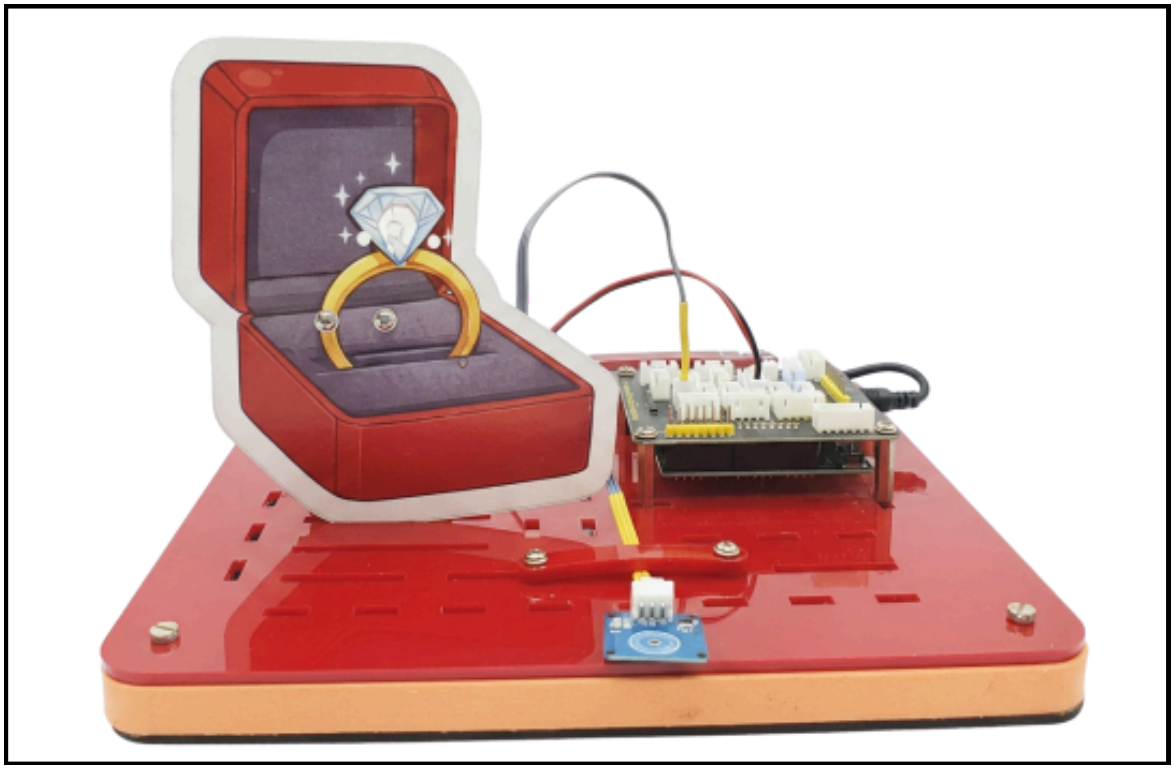


Step 5- Fix the LED to the template and insert the wire through the back

Step 6- Secure the touch sensor properly as shown



Remember- The long leg of the led goes to the red wire and shorter one goes to the black wire



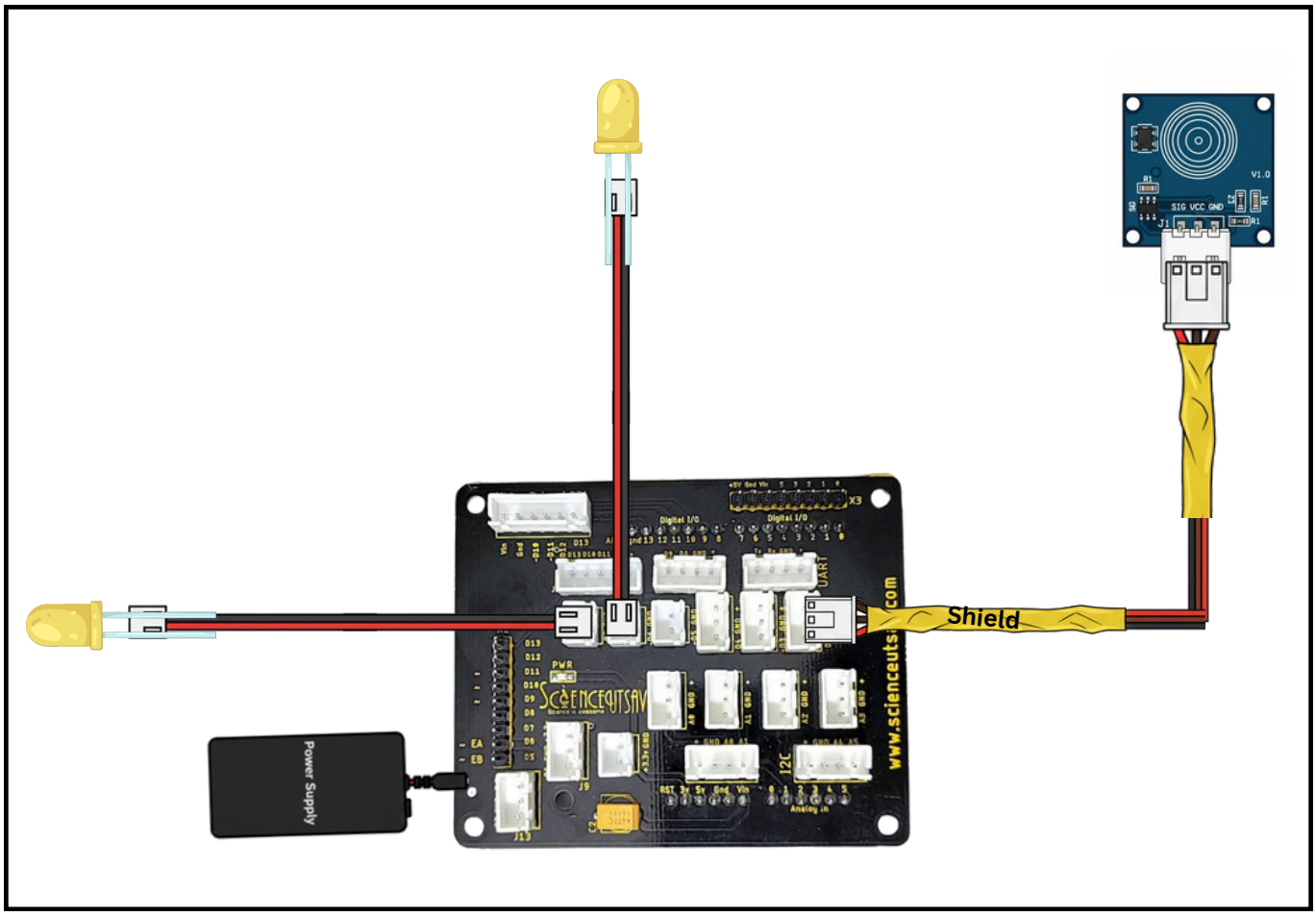
Look at your diamond shine brightly!

Code

```
 /*
ScienceUtsav Arduino Code - Session 6 to 10
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int touchSensor = 2;
const int dipLed = 5;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(dipLed, OUTPUT);
}

void loop() {
  if (digitalRead(touchSensor) == HIGH) digitalWrite(dipLed, HIGH);
  else digitalWrite(dipLed, LOW);
}
```

BEGINNER PROJECT

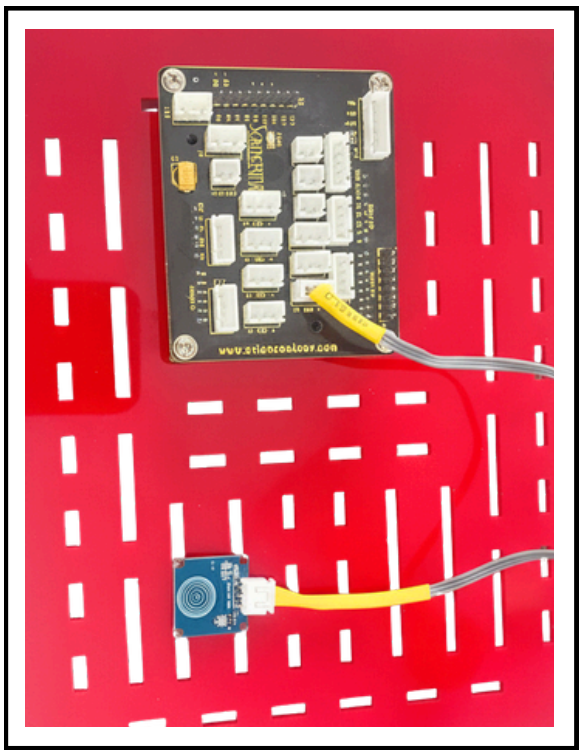
The circuit diagram of the project is given below



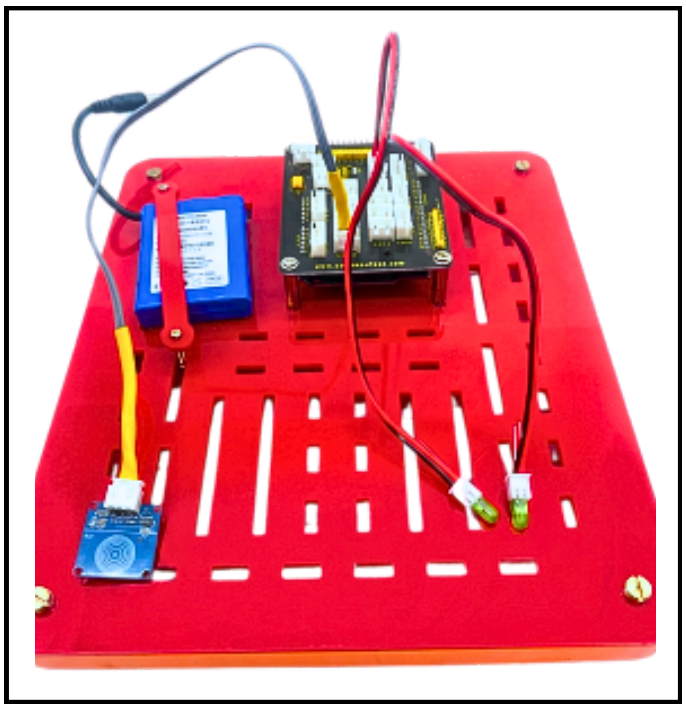
CONNECTIONS:

1. TOUCH SENSOR - D2
2. YELLOW DIP LED - D5, D6

Step 1- Attach the touch sensor to the aurdino as shown

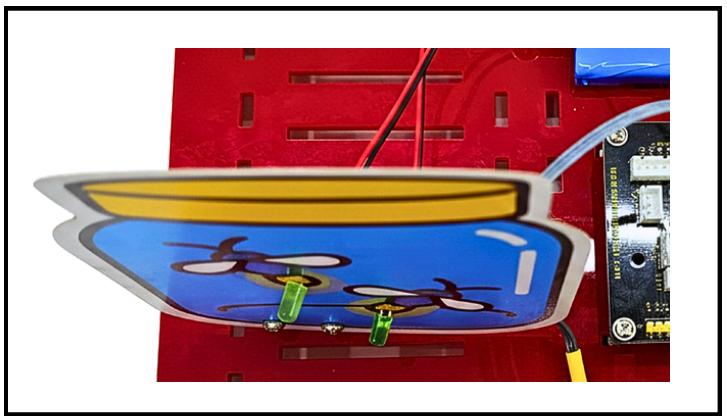


Step 2- Connect the dip LEDs to the aurdino



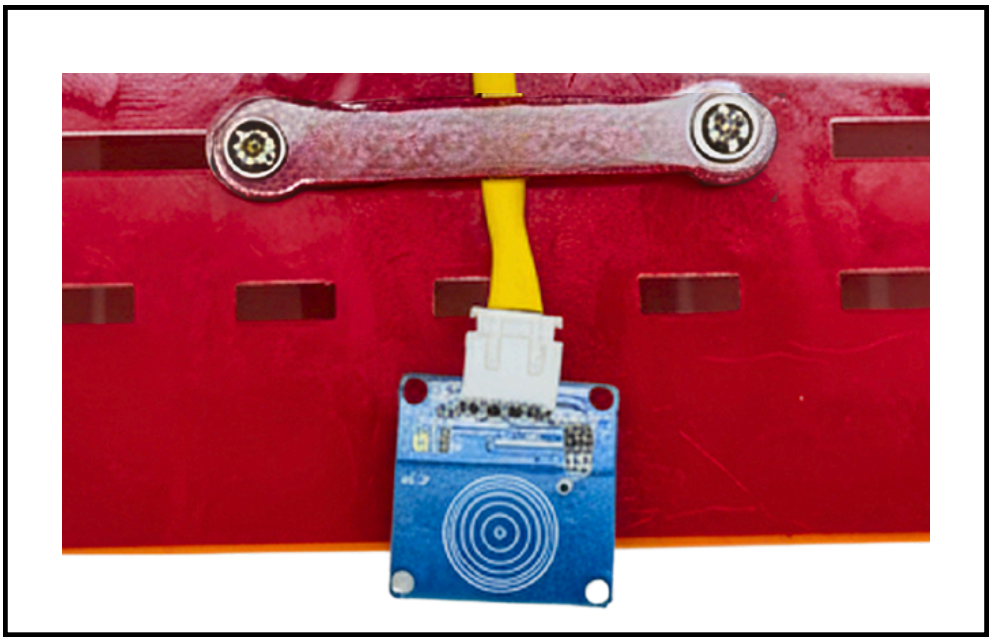
Step 3- Attach the template to the L-clamp and secure it with bolts

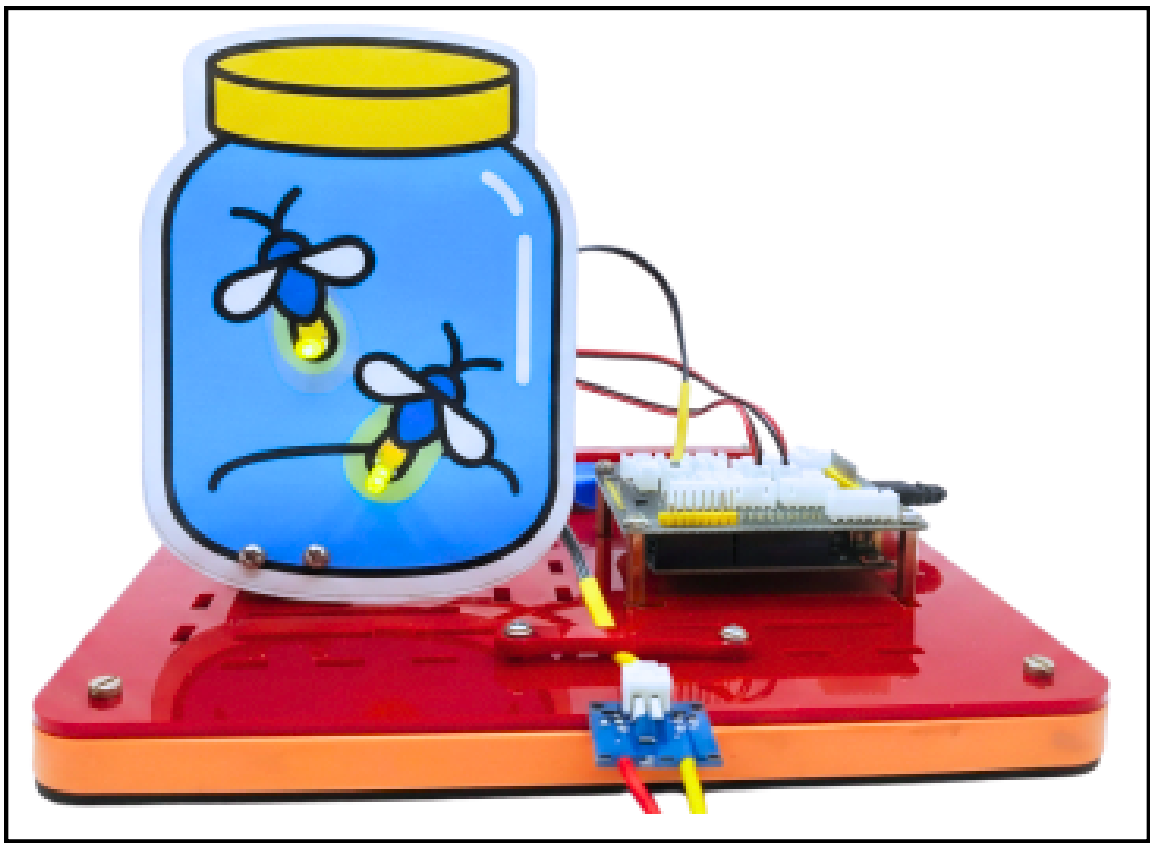
Step 4- Fix the LED to the template and insert the wire through the back



Remember- The long leg of the LED goes to the red wire and the shorter one goes to the black wire

Step 6- Secure the touch sensor properly as shown





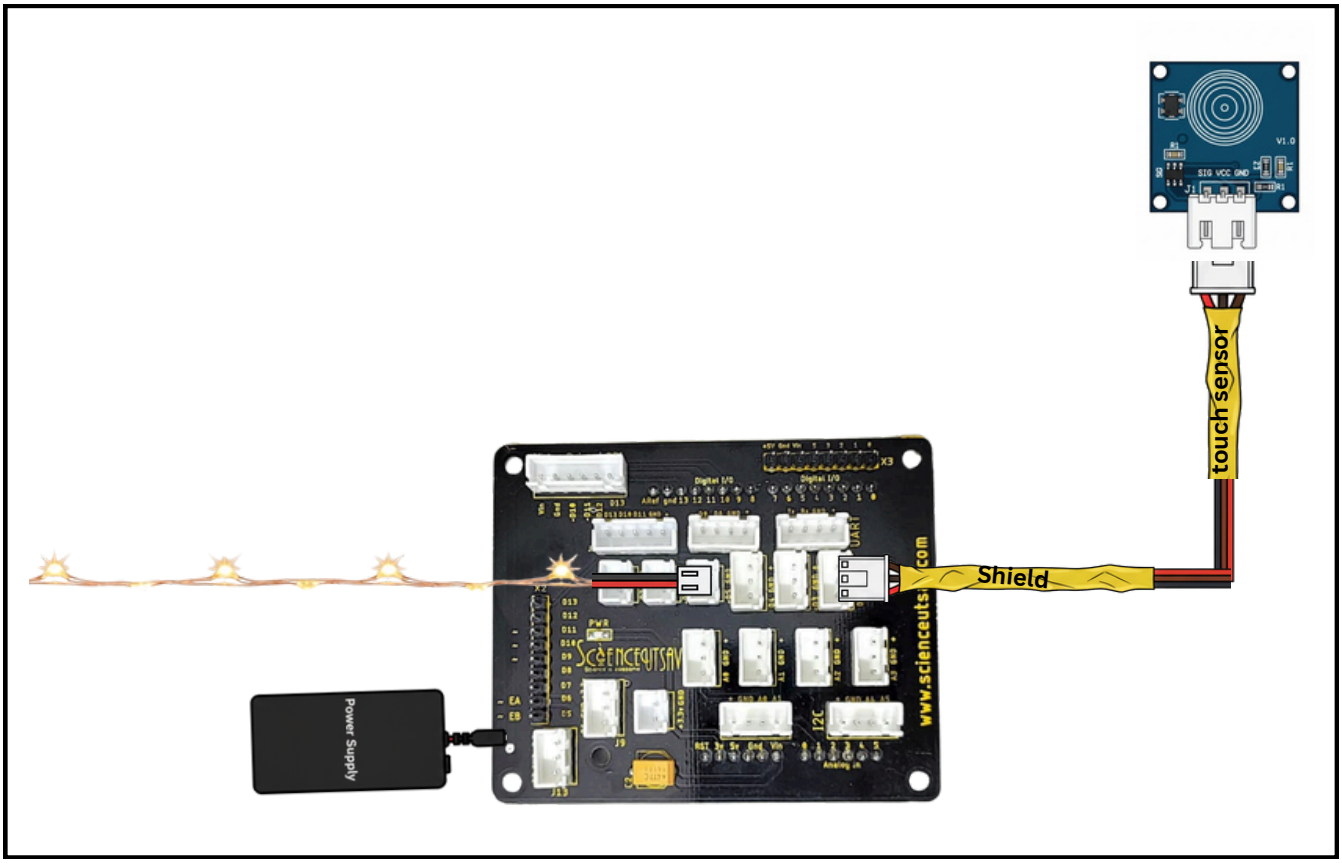
Your fireflies are now ready to glow!

Code

```
/*
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int touchSensor = 2;
const int yellowLed1 = 5;
const int yellowLed2 = 6;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(yellowLed1, OUTPUT);
  pinMode(yellowLed2, OUTPUT);
}

void loop() {
  if (digitalRead(touchSensor) == HIGH) {
    digitalWrite(yellowLed1, HIGH);
    delay(150);
    digitalWrite(yellowLed1, LOW);
    digitalWrite(yellowLed2, HIGH);
    delay(150);
    digitalWrite(yellowLed2, LOW);
  } else {
    digitalWrite(yellowLed1, LOW);
    digitalWrite(yellowLed2, LOW);
  }
}
```

INTERMEDIATE PROJECT

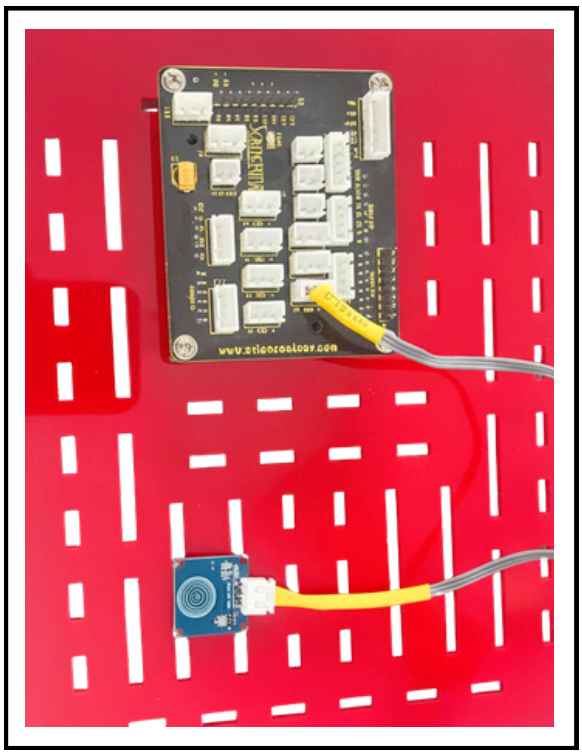
The circuit diagram of the project is given below



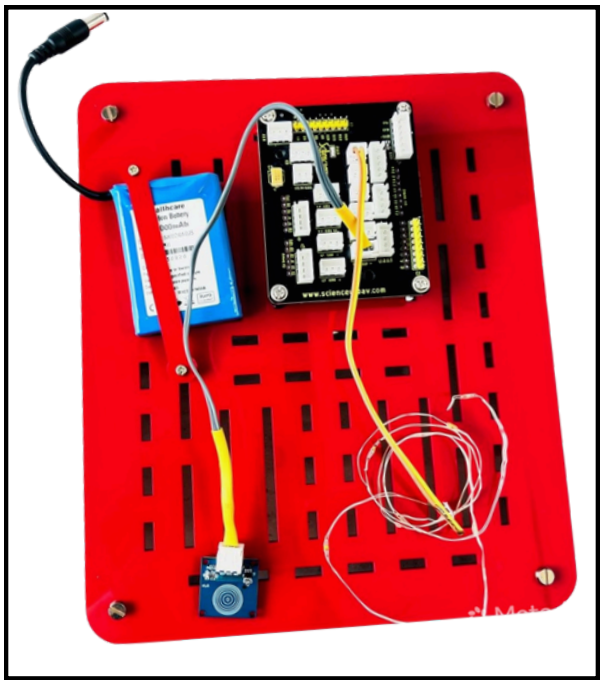
CONNECTIONS:

1. TOUCH SENSOR - D2
2. STRING LED - D5

Step 1- Attach the touch sensor to the aurdino as shown

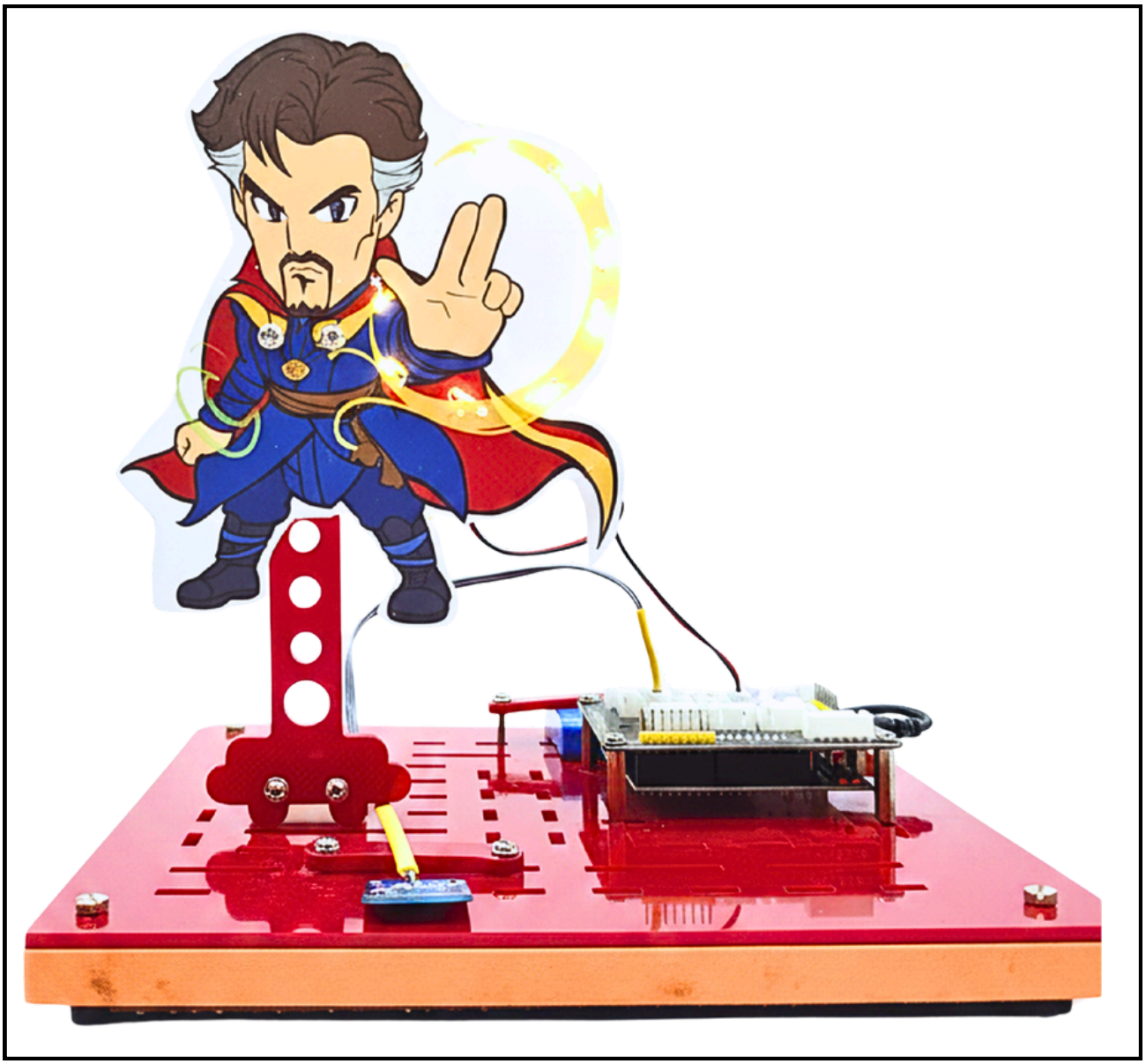


Step 2- Connect the string LEDs to the aurdino



Step 3- Attach the stand and secure the template to it using bolts

Step 4- Place the string led so that the LEDs come to the front and the strings remain in the back as shown



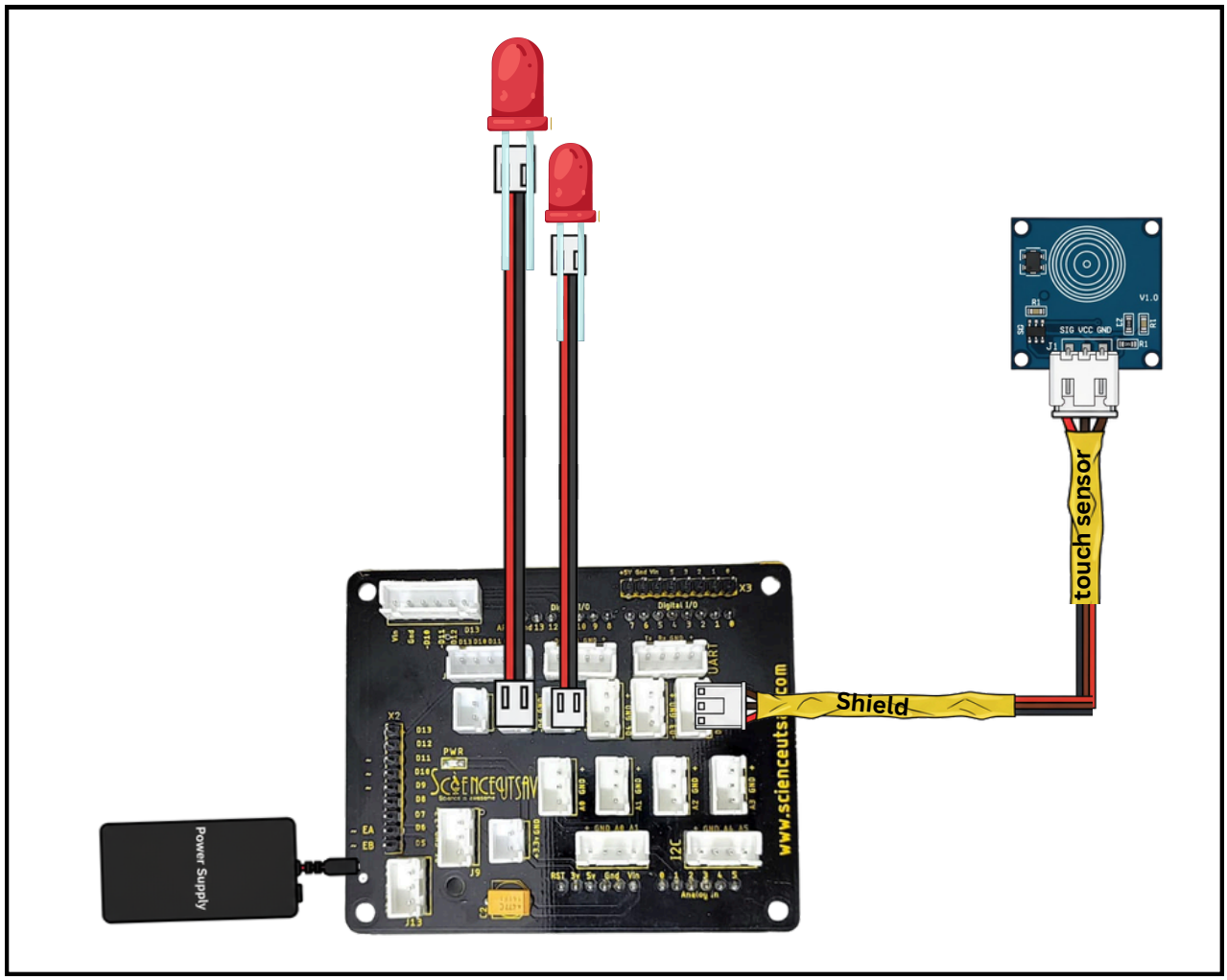
Your superhero is now ready to save the world!

Code

```
/*
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int touchSensor = 2;
const int stringLed = 5;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(stringLed, OUTPUT);
}

void loop() {
  if (digitalRead(touchSensor) == HIGH) {
    for (int i = 0; i < 5; i++) {
      digitalWrite(stringLed, HIGH);
      delay(150);
      digitalWrite(stringLed, LOW);
      delay(150);
    }
  } else {
    digitalWrite(stringLed, LOW);
  }
}
```

INTERMEDIATE PROJECT

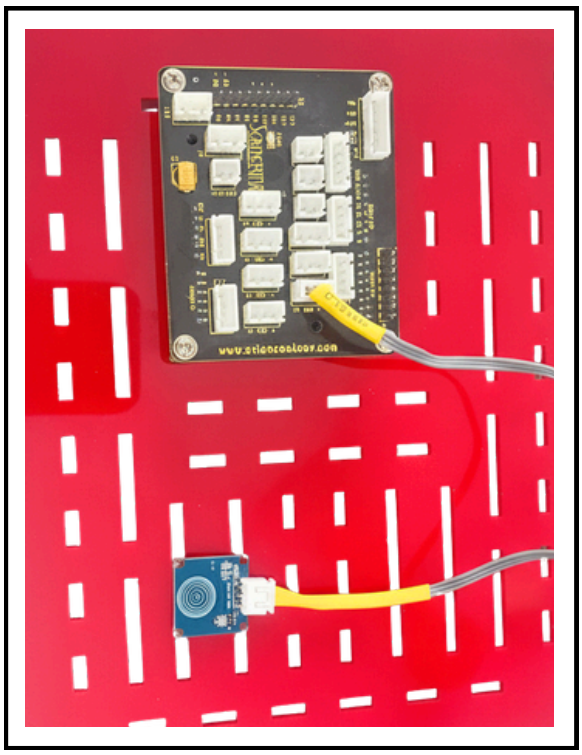
The circuit diagram of the project is given below



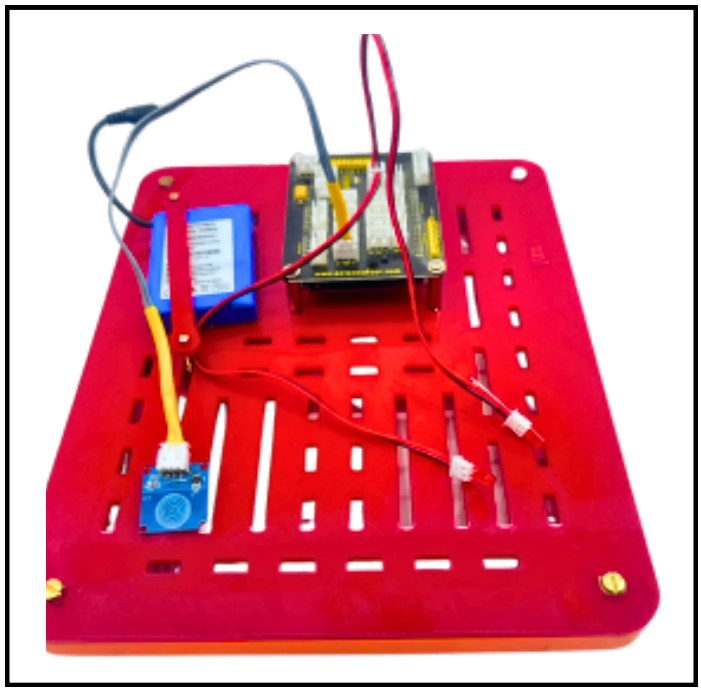
CONNECTIONS:

1. TOUCH SENSOR - D2
2. RED DIP LEDs - D5, D6

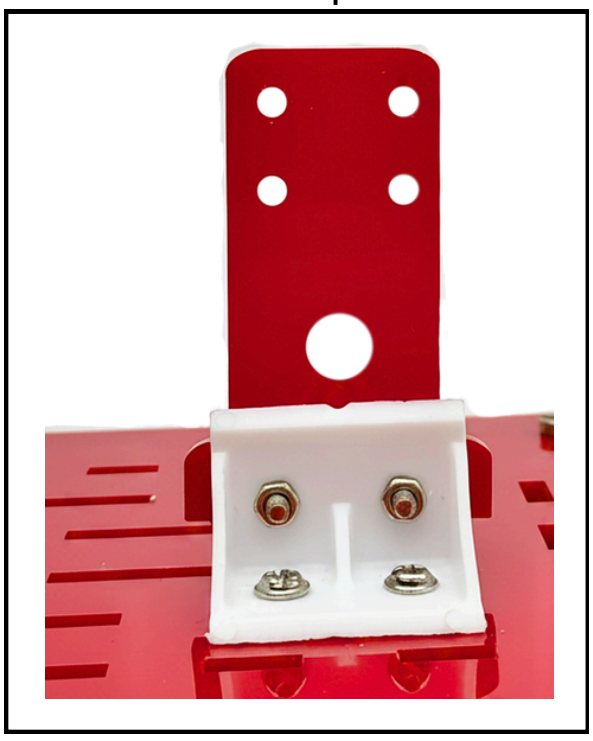
Step 1- Attach the touch sensor to the aurdino as shown



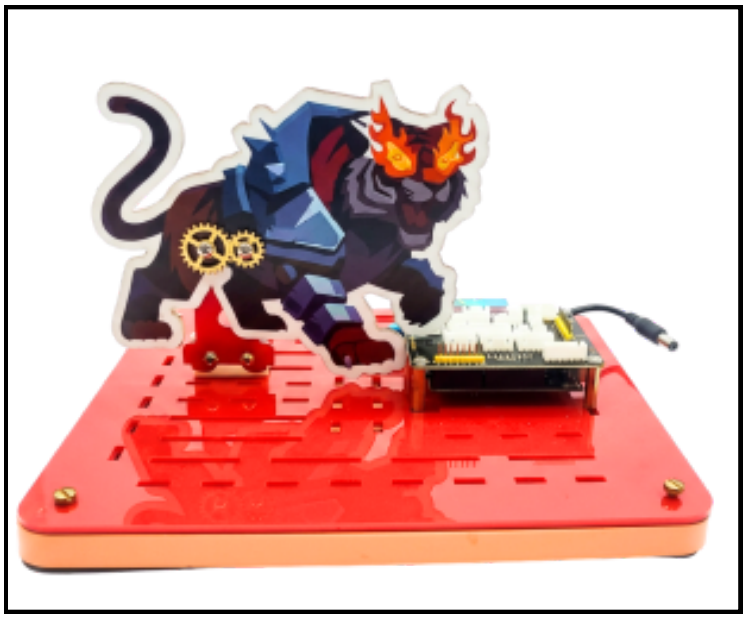
Step 2- Connect two dip LEDs to the aurdino



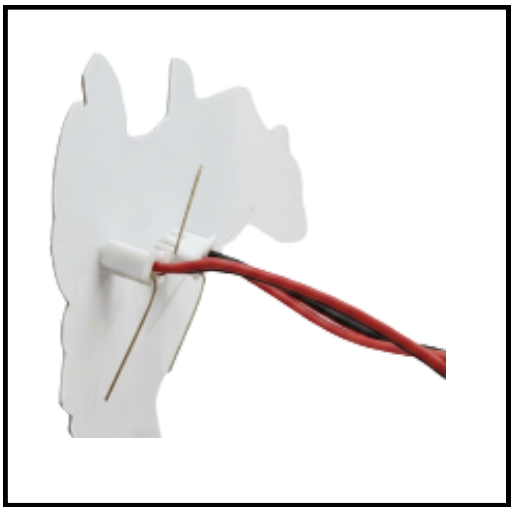
Step 3- Attach the stand as shown with the help of L-clamp



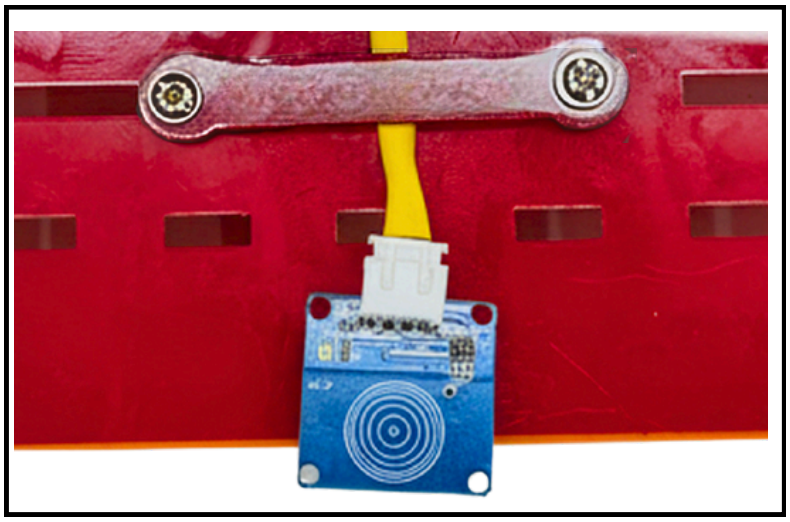
Step 4- Attach the template and secure with bolts



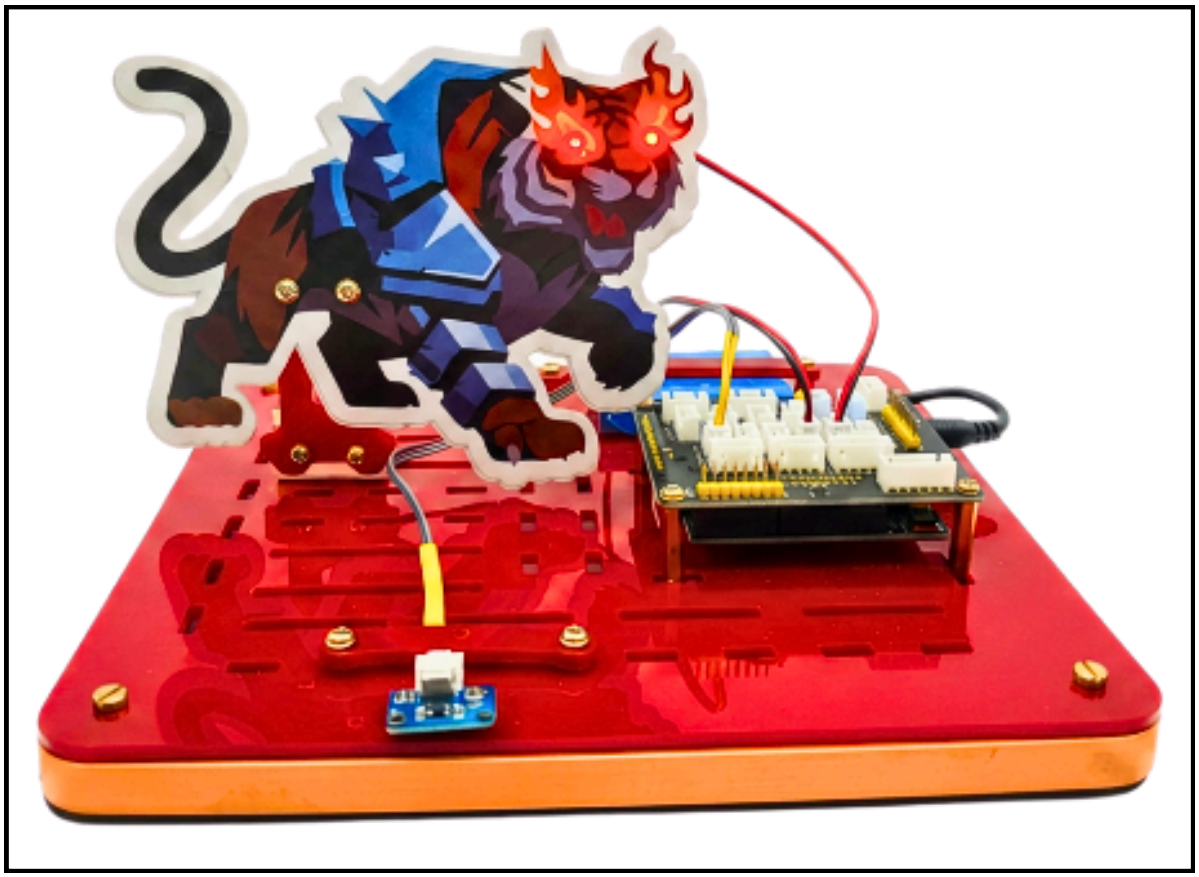
Step 5- Fix the LED to the template and insert the wire through the back



Step 6- Secure the touch sensor properly as shown



Remember- The long leg of the LED goes to the red wire and the shorter one goes to the black wire



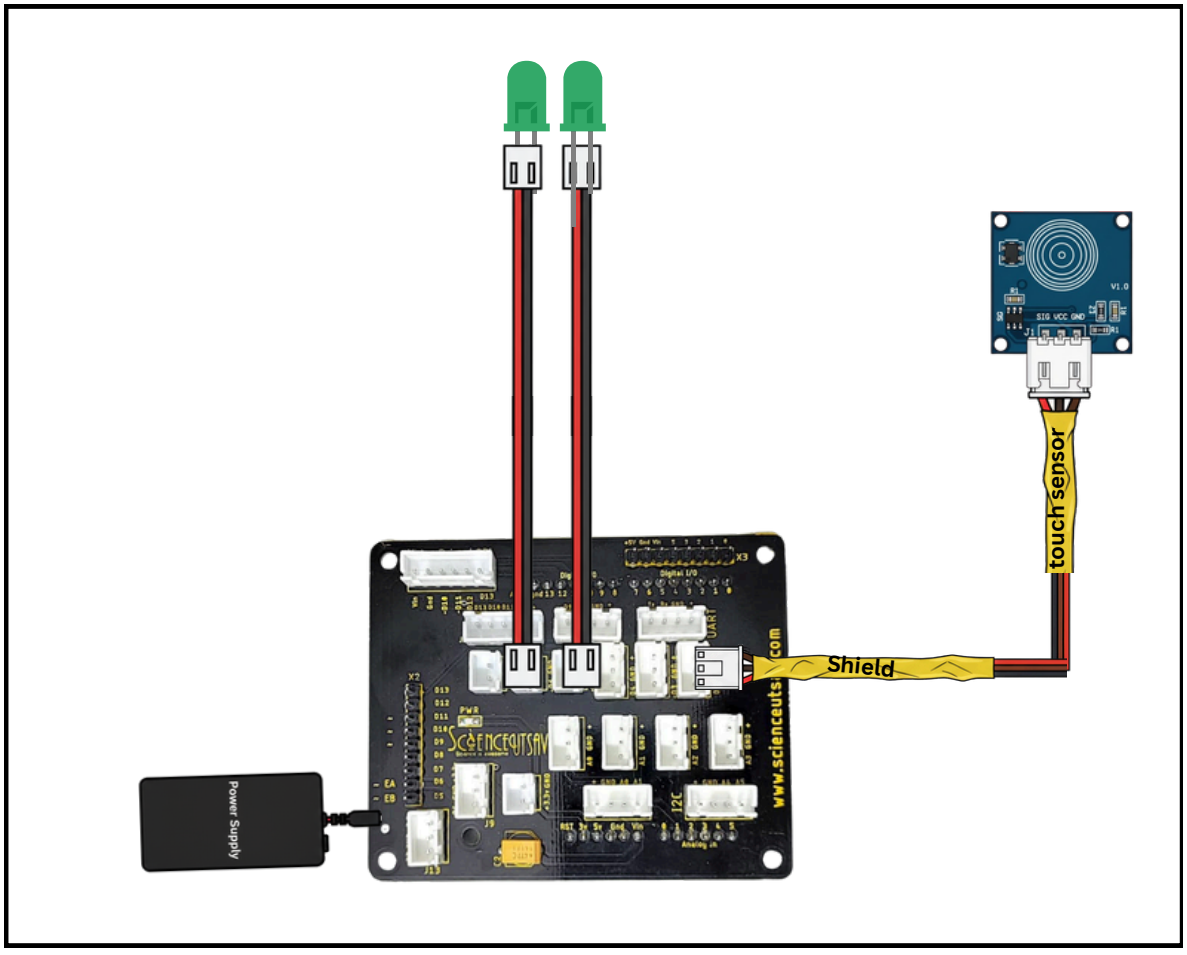
Your tiger is now ready to roar!

Code

```
/*
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int touchSensor = 2;
const int redLed1 = 5;
const int redLed2 = 6;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(redLed1, OUTPUT);
  pinMode(redLed2, OUTPUT);
}

void loop() {
  if (digitalRead(touchSensor) == HIGH) {
    digitalWrite(redLed1, HIGH);
    digitalWrite(redLed2, HIGH);
  } else {
    digitalWrite(redLed1, LOW);
    digitalWrite(redLed2, LOW);
  }
}
```

ADVANCE PROJECT

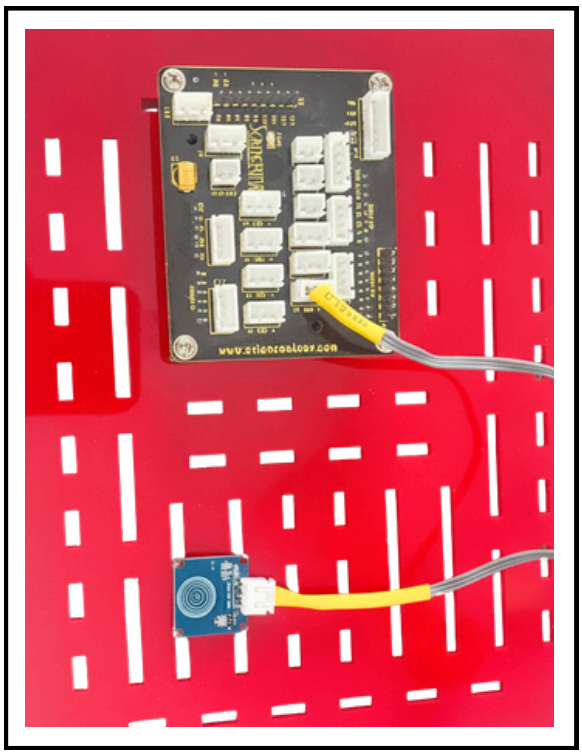
The circuit diagram of the project is given below



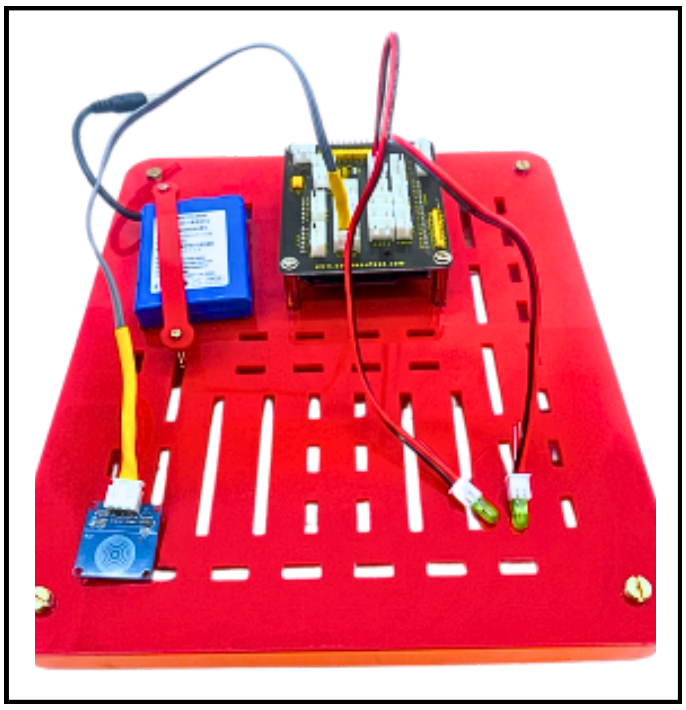
CONNECTIONS:

1. TOUCH SENSOR - D2
2. GREEN DIP LEDs - D5, D6

Step 1- Attach the touch sensor to the aurdino as shown

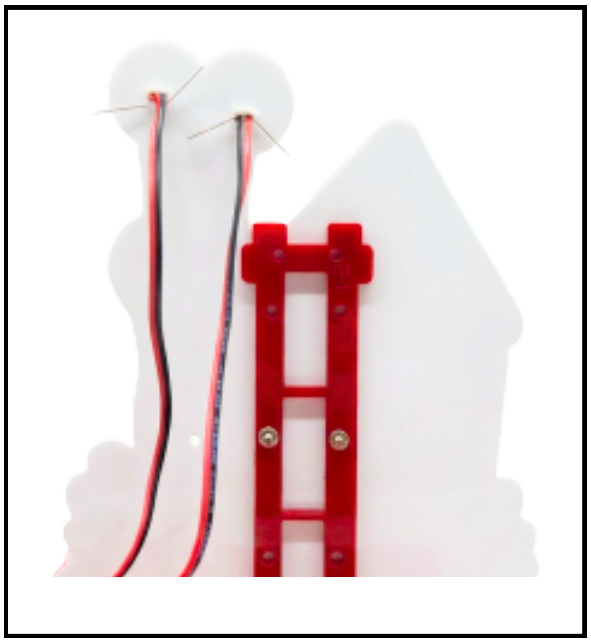


Step 2- Connect the dip LEDs to the aurdino



Step 3- Attach the stand and secure the template to it using bolts

Step 4- Push the dip LEDs through the holes and insert the wires from the back



Remember- The long leg of the LED goes to the red wire and the shorter one goes to the black wire

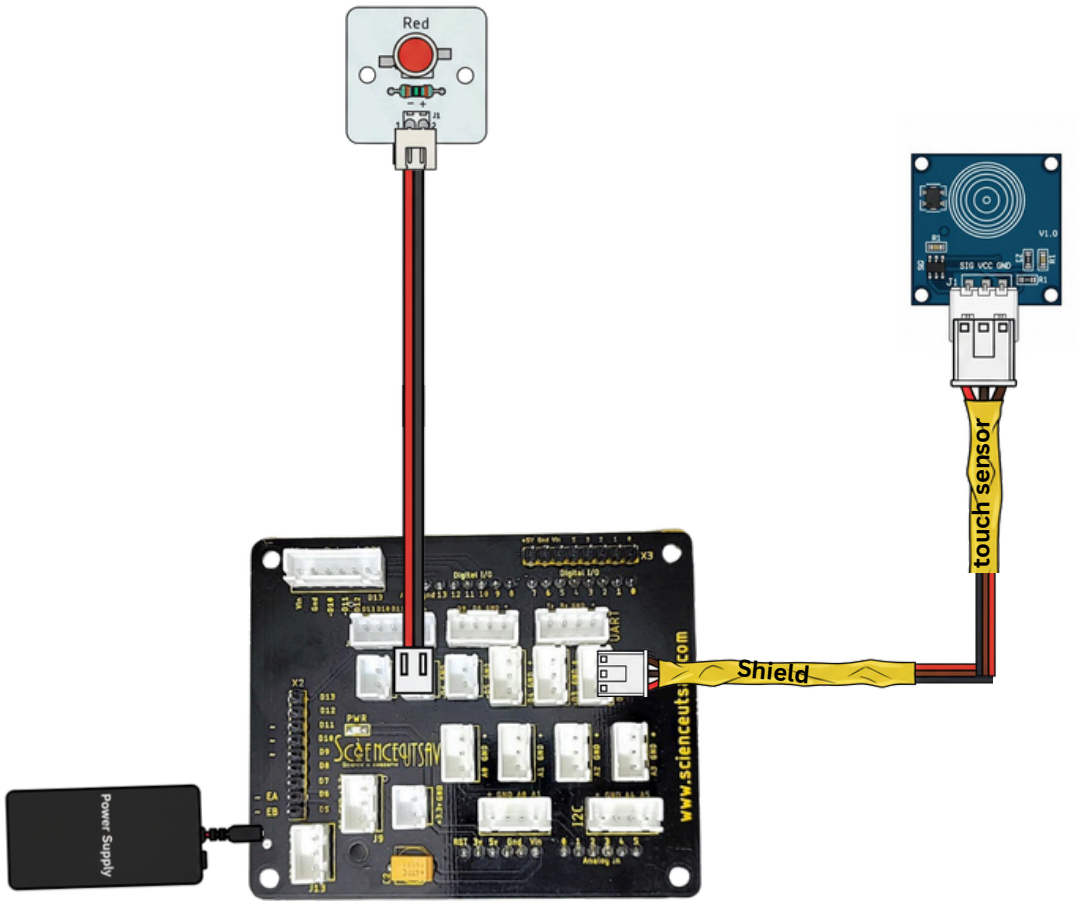
Your snail is ready!

Code

```
/*
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int touchSensor = 2;
const int dipLed1 = 5;
const int dipLed2 = 6;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(dipLed1, OUTPUT);
  pinMode(dipLed2, OUTPUT);
}

void loop() {
  if (digitalRead(touchSensor) == HIGH) {
    digitalWrite(dipLed1, HIGH);
    delay(400);
    digitalWrite(dipLed2, HIGH);
    delay(400);
    digitalWrite(dipLed1, LOW);
    digitalWrite(dipLed2, LOW);
    delay(400);
  } else {
    digitalWrite(dipLed1, LOW);
    digitalWrite(dipLed2, LOW);
  }
}
```

ADVANCE PROJECT

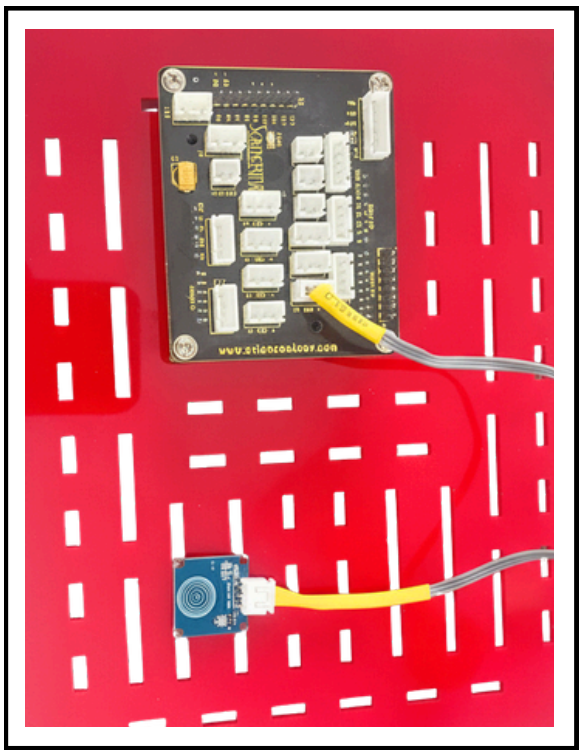
The circuit diagram of the project is given below



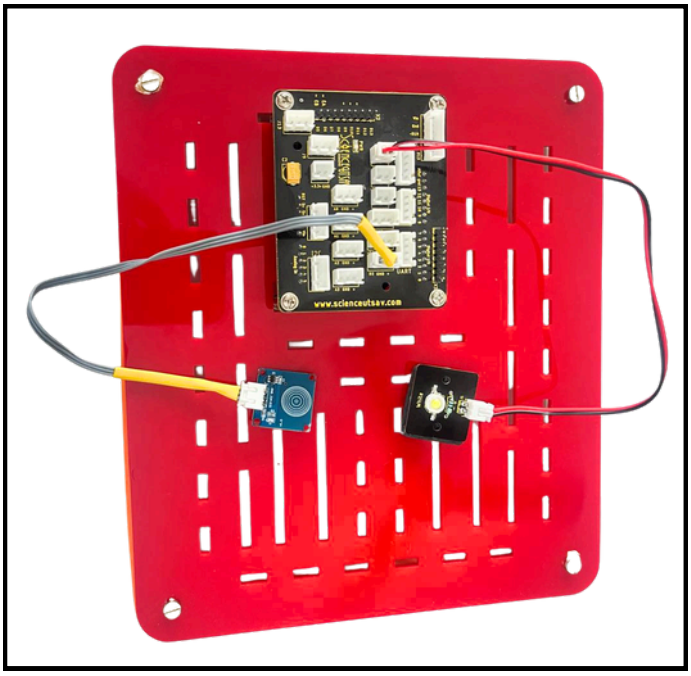
CONNECTIONS:

1. TOUCH SENSOR - D2
2. 1W RED LED - D6

Step 1- Attach the touch sensor to the aurdino as shown

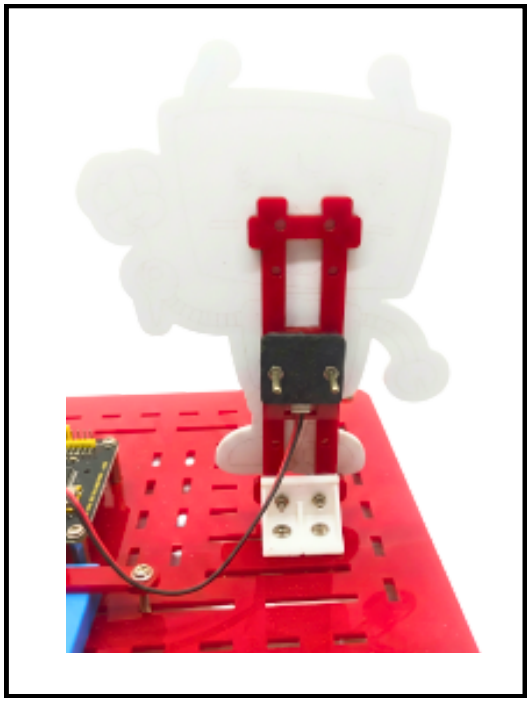


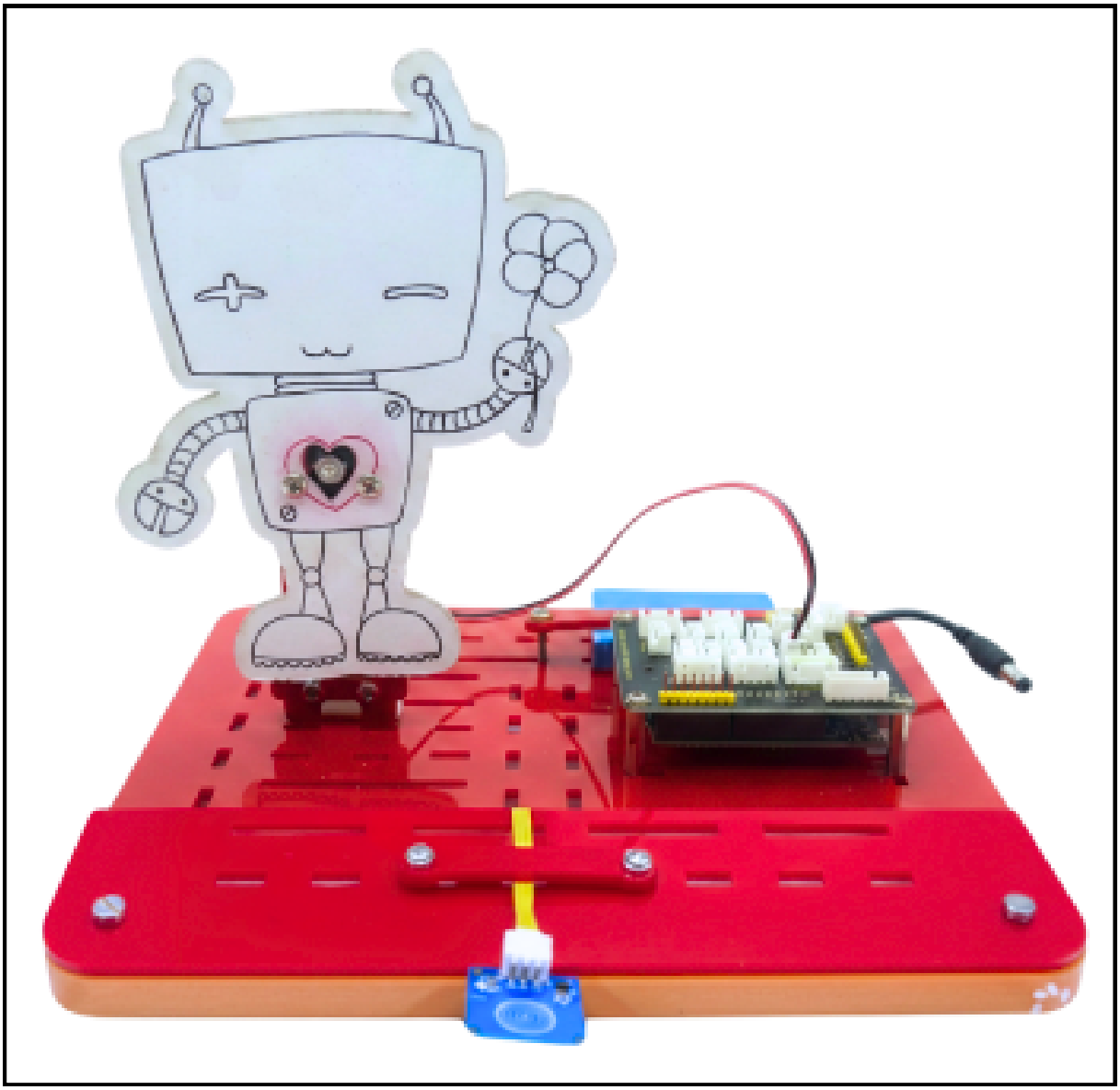
Step 2- Connect the LED to the aurdino



Step 3- Attach the stand. secure the template to it using the bolts

Step 4- Fix the LED to the stand as shown





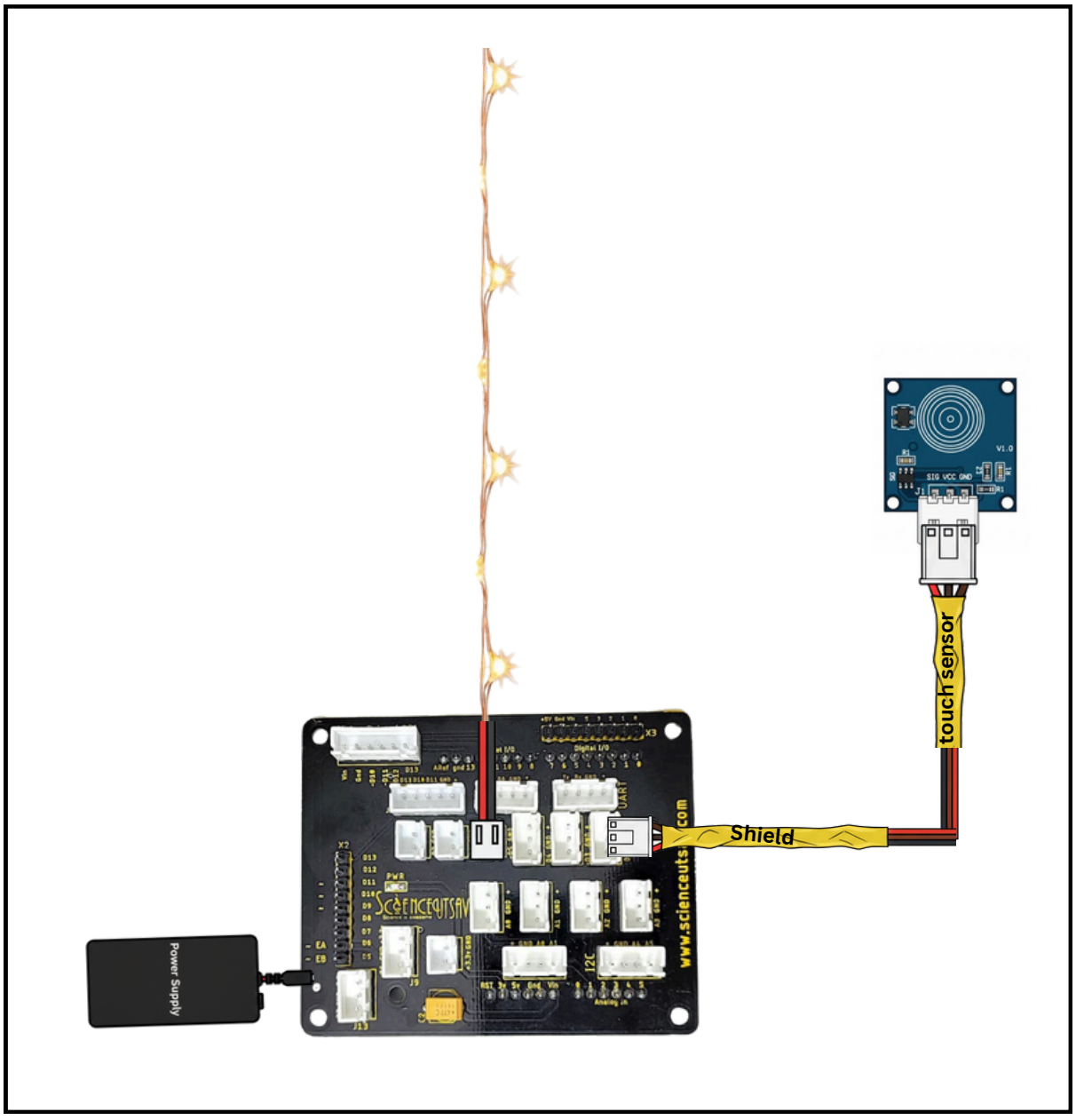
Your robo is ready!

Code

```
/*
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int touchSensor = 2;
const int led = 6;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(led, OUTPUT);
}

void loop() {
  if (digitalRead(touchSensor) == HIGH) digitalWrite(led, HIGH);
  else digitalWrite(led, LOW);
}
```

EXPERT PROJECT

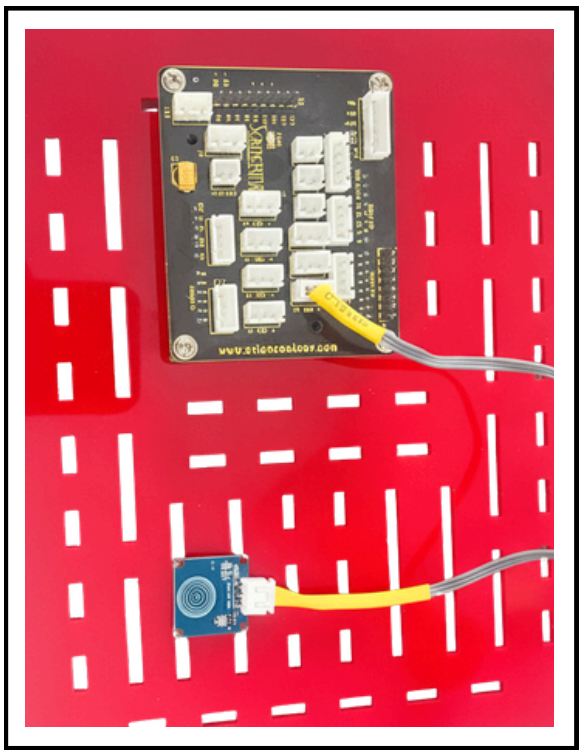
The circuit diagram of the project is given below



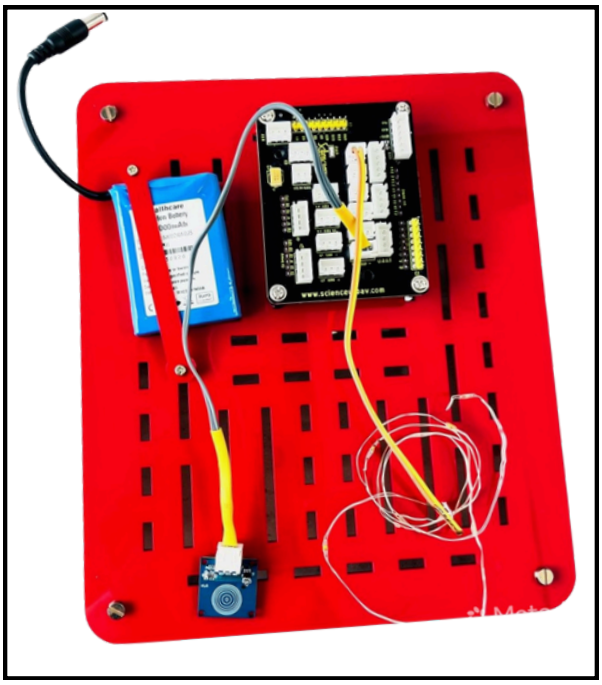
CONNECTIONS:

1. TOUCH SENSOR - D2
2. STRING LED - D5

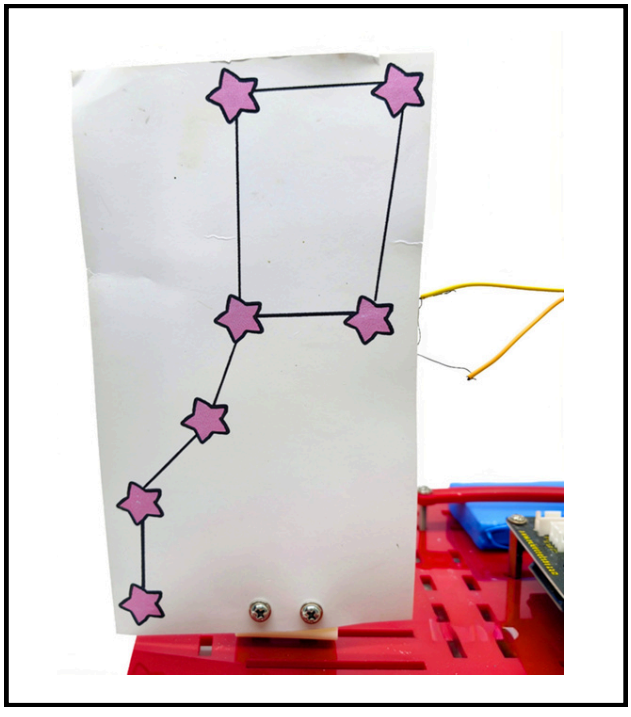
Step 1- Attach the touch sensor to the aurdino as shown



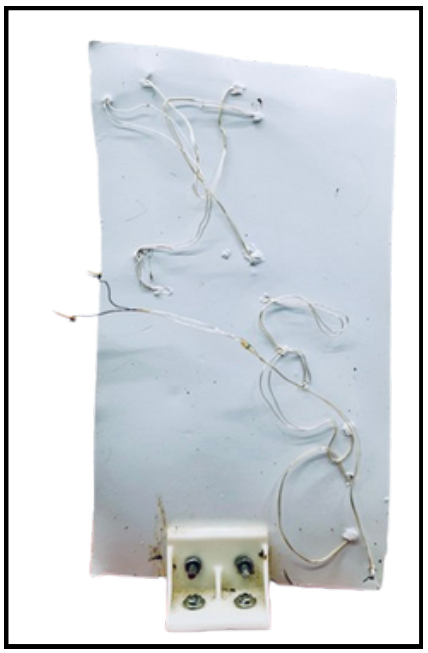
Step 2- Connect the string LEDs to the aurdino



Step 3- Attach the template to the L-clamp and fix it to the board



Step 4- String the string led through the holes so that the LEDs come to the front and the strings remain in the back as shown



Enjoy the twinkling of the Big Dipper!

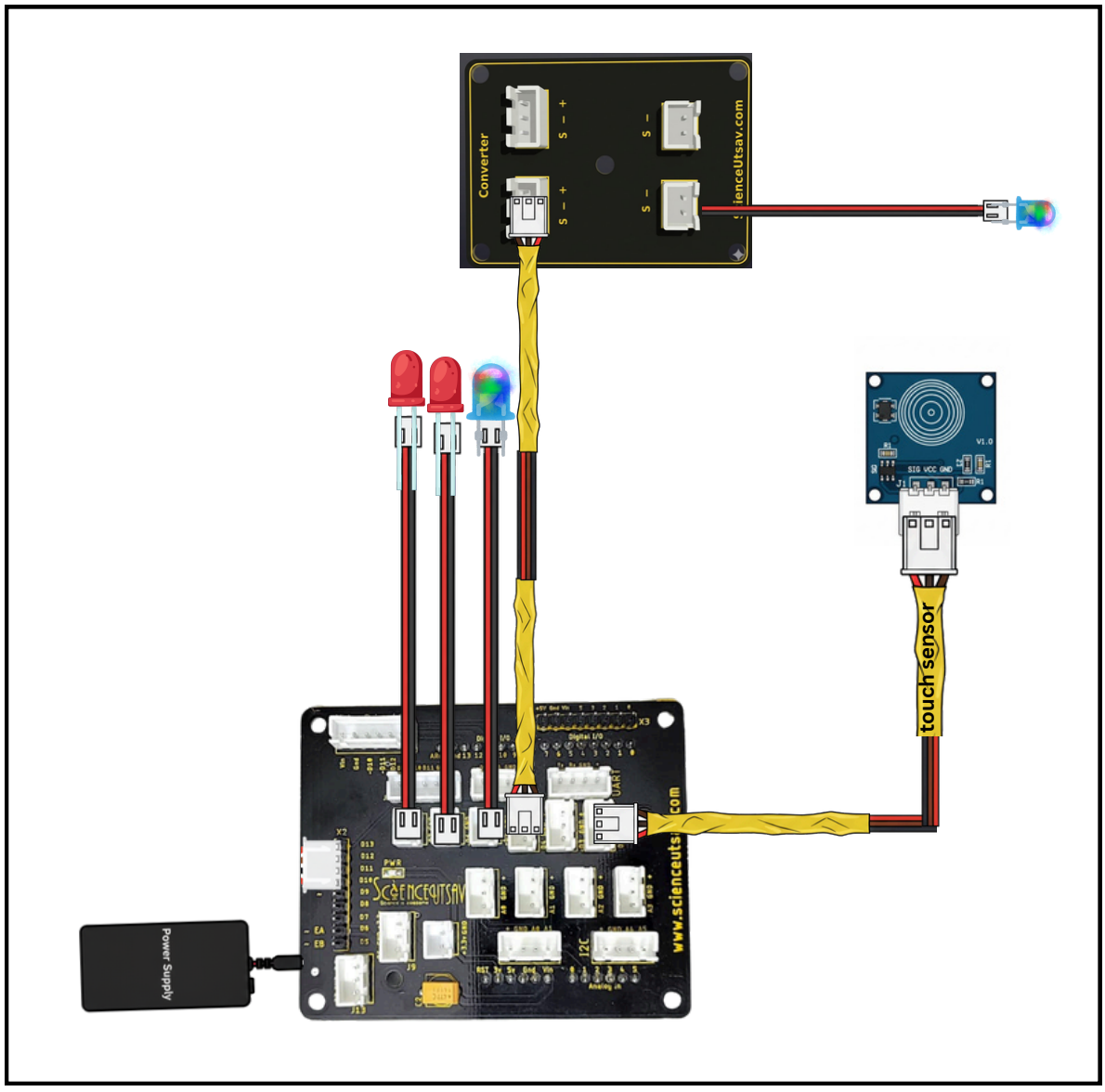
Code

```
/*
ScienceUtsav Arduino Code - Touch to Change Pattern
Touch Sensor = D2
String LED = D5
*/

const int touchSensor = 2;
const int stringLed = 5;

int pattern = 0;
bool lastTouchState = LOW;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(stringLed, OUTPUT);
}

void loop() {
  bool touchState = digitalRead(touchSensor);

  // Change pattern on every new touch
  if (touchState == HIGH && lastTouchState == LOW) {
    pattern++;

    if (pattern > 3) {
      pattern = 0;
    }

    delay(300);
  }
```

```
lastTouchState = touchState;

  // Pattern 0: LED OFF
  if (pattern == 0) {
    analogWrite(stringLed, 0);
  }

  // Pattern 1: Fade in and fade out
  if (pattern == 1) {
    for (int brightness = 0; brightness <= 255; brightness += 5) {
      analogWrite(stringLed, brightness);
      delay(20);
    }

    for (int brightness = 255; brightness >= 0; brightness -= 5) {
      analogWrite(stringLed, brightness);
      delay(20);
    }
  }

  // Pattern 2: Slow blinking
  if (pattern == 2) {
    analogWrite(stringLed, 255);
    delay(500);
    analogWrite(stringLed, 0);
    delay(500);
  }

  // Pattern 3: Fast blinking
  if (pattern == 3) {
    analogWrite(stringLed, 255);
    delay(100);
    analogWrite(stringLed, 0);
    delay(100);
  }
}
```

EXPERT PROJECT

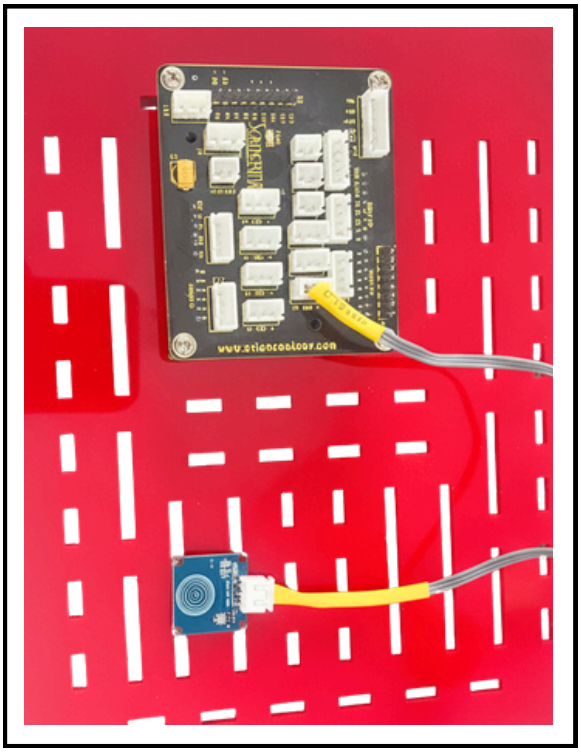
The circuit diagram of the project is given below



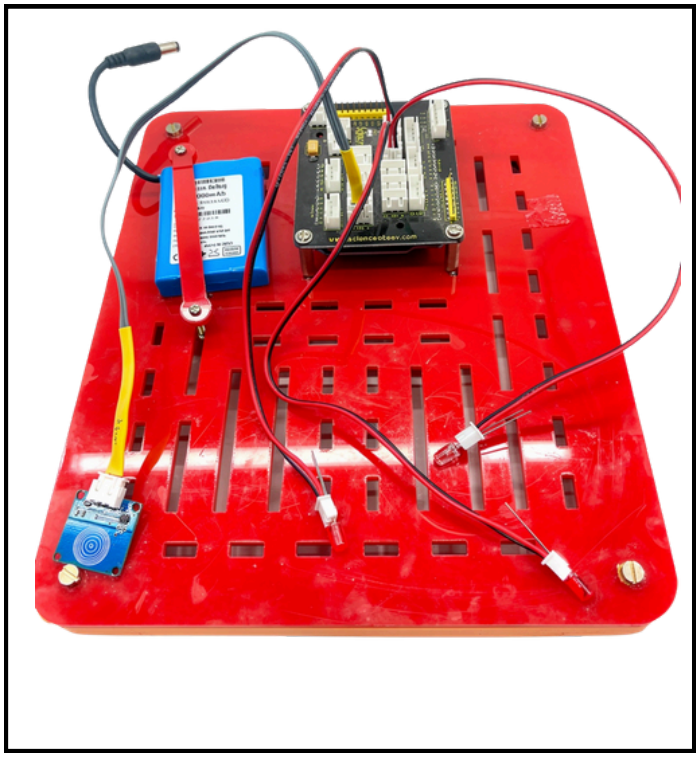
CONNECTIONS:

1. TOUCH SENSOR - D2
2. RED DIP LEDs - D7, D7
3. MULTICOLOUR - D5
4. CONVERTER - D4
5. CONVERTER - MULTICOLOUR LED

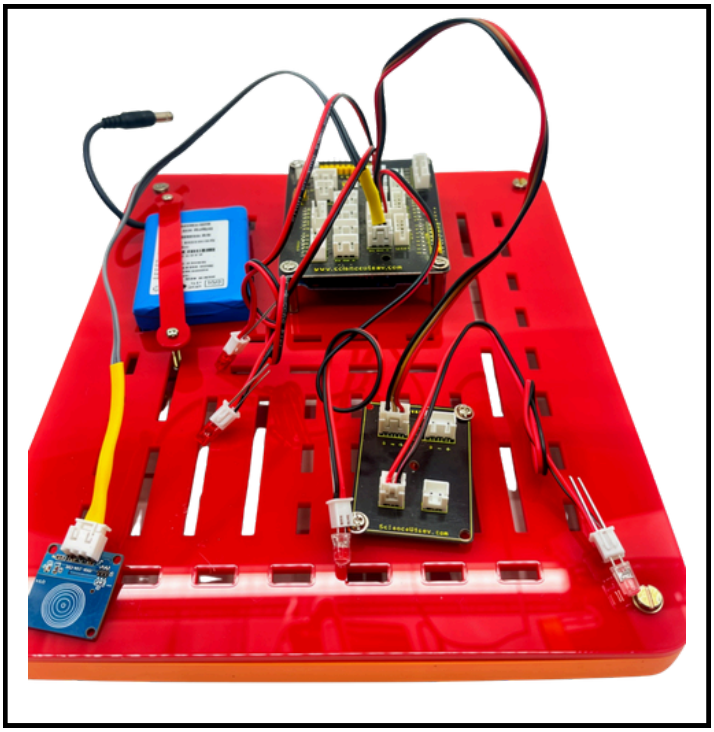
Step 1- Attach the touch sensor to the aurdino as shown



Step 2- Connect three dip LEDs to the board

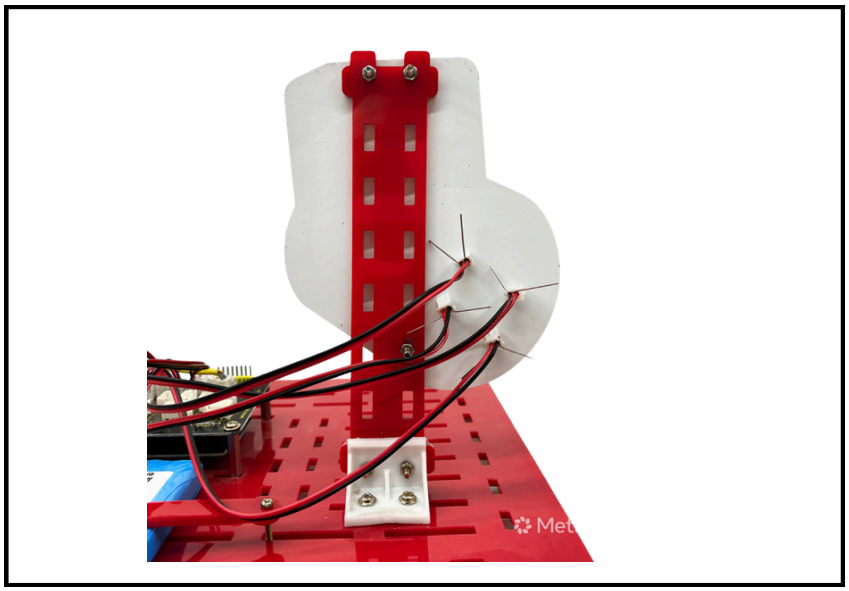


Step 3- Connect a fourth LED using a converter

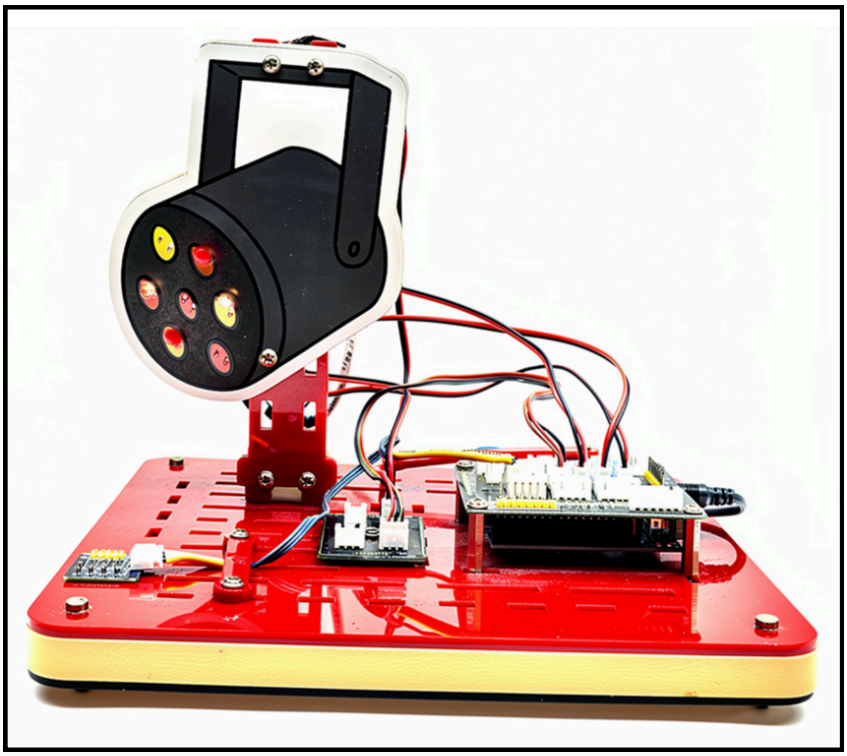


Step 4- Fix the stand and the template and secure it with bolts

Step 5- Insert the LEDs through the front and the wires from the back



Remember- The long leg of the LED goes to the red wire and the shorter one goes to the black wire



Have fun with your Disco Light!

Code

```
/*
ScienceUtsav Arduino Code
Touch Sensor with 4 LED Patterns

Touch Sensor = D2
LEDs = D4, D5, D6, D7
*/

const int touchSensor = 2;
const int leds[] = {4, 5, 6, 7};

int touchCount = 0;
bool lastTouchState = LOW;

void setup() {
  pinMode(touchSensor, INPUT);

  for (int i = 0; i < 4; i++) {
    pinMode(leds[i], OUTPUT);
  }
}

void allLeds(bool state) {
  for (int i = 0; i < 4; i++) {
    digitalWrite(leds[i], state);
  }
}

void loop() {
  bool touchState = digitalRead(touchSensor);

  if (touchState == HIGH && lastTouchState == LOW) {
    touchCount = (touchCount + 1) % 5;
    delay(300);
  }
```

```
  lastTouchState = touchState;

  switch (touchCount) {

    case 0:   // All LEDs OFF
      allLeds(LOW);
      break;

    case 1:   // All LEDs ON
      allLeds(HIGH);
      break;

    case 2:   // One by one blinking
      for (int i = 0; i < 4; i++) {
        allLeds(LOW);
        digitalWrite(leds[i], HIGH);
        delay(200);
      }
      break;

    case 3:   // Alternate LEDs
      digitalWrite(leds[0], HIGH);
      digitalWrite(leds[1], LOW);
      digitalWrite(leds[2], HIGH);
      digitalWrite(leds[3], LOW);
      delay(300);

      digitalWrite(leds[0], LOW);
      digitalWrite(leds[1], HIGH);
      digitalWrite(leds[2], LOW);
      digitalWrite(leds[3], HIGH);
      delay(300);
      break;

    case 4:   // Fast blinking
      allLeds(HIGH);
      delay(150);
      allLeds(LOW);
      delay(150);
      break;
  }
}
```